

Thermal Diffusivity Lab

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February 24, 2026

Abstract

Thermal Diffusivity is a material property that describes the mitigation speed of heat within the material itself. In this experiment, the thermal diffusivity of lab-used silicone rubber is determined by measuring a time delay for heat response between alternating ice/hot water bath, for chosen angular frequencies. Using a non-linear fit, the thermal diffusivity of silicone rubber is determined to be $m_{exp} = (4.2 \pm 0.3) \times 10^{-8} [m^2/s]$, demonstrating the long time period required for heat to mitigate through silicone rubber.

Introduction

Thermal Diffusivity

The propagation of thermal energy through an object (also known as conduction) is governed by the partial differential equation:

$$\frac{\partial T}{\partial t} = m \nabla^2 T \quad (1)$$

where $T [K]$ is the temperature, $t [s]$ is time, and $m [m^2/s]$ is the object's thermal diffusivity which, in general, is given as a function of its other thermal properties:

$$m = \frac{k}{\rho c_p} \quad (2)$$

where $k [W/mK]$ is the thermal conductivity, $\rho [kg/m^3]$ is the density, $c_p [J/kgK]$ is the specific heat capacity.

In the case of an object with cylindrical symmetry, which is a rubber cylindrical shell for this lab, Equation 1 becomes:

$$\frac{\partial T}{\partial t} = m \left(\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} \right) \quad (3)$$

assuming that the rubber cylindrical shell is long ($z \gg r$).

Solving Equation 3 involves using the method of separation of variables by substituting $T(r, t) = R(r)e^{i\omega t}$, with ω [rad/s] as the angular frequency of an alternating temperature applied to the surface of the rubber (consistent to the experimental procedure). The solution obtained is J_0 , a Bessel function of order zero, where the phase difference in temperature ϕ [rad] at some point in the rubber cylindrical shell from the temperature at the centre can be expressed as:

$$\tan \phi = \frac{\frac{x^2}{2^2} - \frac{x^6}{2^2 4^2 6^2} + \dots}{1 - \frac{x^4}{2^2 4^2} + \dots} \quad (4)$$

and x is a dimensionless variable such that $x^2 = r^2 \omega / m$.

In addition, the time delay Δt [s] between an abrupt change in temperature at the surface of the rubber cylindrical shell and the closest maximum or minimum temperature reached at the inner radius of the rubber cylindrical shell can be expressed as:

$$\phi(r_1) - \phi(r_2) = \omega \Delta t \quad (5)$$

where r_1 and r_2 [m] are the inner and outer radii of the rubber cylindrical shell respectively, and ϕ is the phase difference as a function of the radius.

Therefore, taking Equation 5 and substituting Equation 4 for ϕ terms, the time delay as a function of ω can be approximated as:

$$\Delta t(\omega) = \frac{1}{\omega} \left[\arctan\left(\frac{\frac{1}{4}\left(\frac{r_1^2 \omega}{m}\right) - \frac{1}{2304}\left(\frac{r_1^2 \omega}{m}\right)^3}{1 - \frac{1}{64}\left(\frac{r_1^2 \omega}{m}\right)^2}\right) - \arctan\left(\frac{\frac{1}{4}\left(\frac{r_2^2 \omega}{m}\right) - \frac{1}{2304}\left(\frac{r_2^2 \omega}{m}\right)^3}{1 - \frac{1}{64}\left(\frac{r_2^2 \omega}{m}\right)^2}\right) \right] \quad (6)$$

Uncertainties and Uncertainty Propagation

Uncertainties are a measure of how precise the measurement or calculation of a variable is. The two types of uncertainties are instrumental and statistical uncertainty, where the instrumental uncertainty is given by measurement tools and procedures, and the statistical uncertainty is the standard deviation of repeated measurements defined as:

$$\sigma_x = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (\bar{x} - x_i)^2} \quad (7)$$

where N is the number of data points, $\bar{x}$ is the average value of the data points, and x_i is the i th data point. [1]

The instrumental and statistical uncertainties of a variable measured repeatedly can be combined to provide an overall uncertainty using:

$$\sigma = \sqrt{\sigma_{ins}^2 + \sigma_{stdev}^2} \quad (8)$$

where σ is the overall uncertainty, σ_{ins} is the instrumental uncertainty, and σ_{stdev} is the statistical uncertainty, as is consistent with [2].

Furthermore, in order to propagate uncertainties in calculations, the general formula for uncertainty propagation is needed:

$$\Delta z = \sqrt{\sum_{i=1}^N \left(\frac{\partial f}{\partial x_i} \Delta x_i \right)^2} \quad (9)$$

where Δz is the uncertainty of the result, f is the function used to calculate the result, N is the number of parameters in the function, x_i is the i th parameter of f , and Δx_i is the uncertainty associated with the i th parameter.

From Equation 9, the uncertainty propagation formula for addition and subtraction can be derived:

$$\sigma_c = \sqrt{\sigma_a^2 + \sigma_b^2} \quad (10)$$

where σ denotes the uncertainty of the associated variables such that $c = a \pm b$.

Angular Frequency

The angular frequency ω [rad/s] of an alternating temperature is dependent on its period P [s] through the formula:

$$\omega = 2\pi/P \quad (11)$$

Using Equation 9, the uncertainty propagation formula from P to ω is:

$$\Delta\omega = \frac{2\pi}{P^2} \Delta P \quad (12)$$

where Δ denotes the uncertainty of the associated variable.

Experimental Method

Materials

- Hot Plate
- Stand and Clamps
- 2 1000 mL Beakers
- Thermometer
- Silicone Rubber Cylindrical Shell
- Ice

- Metal Scissor lift
- Timer
- Ruler

Procedures

1. Measure the inner and outer diameter of the rubber cylindrical shell using a ruler.
2. Fill one beaker with water (Approximately 900 mL) and place on a hot plate, turn the plate to high and wait for it to boil.
3. Fill the other beaker with water and ice (Approximately a combined 900 mL) and place it on the metal scissor lift.
4. Adjust the ice bath to the same height as the boiling beaker, using the metal scissor lift.

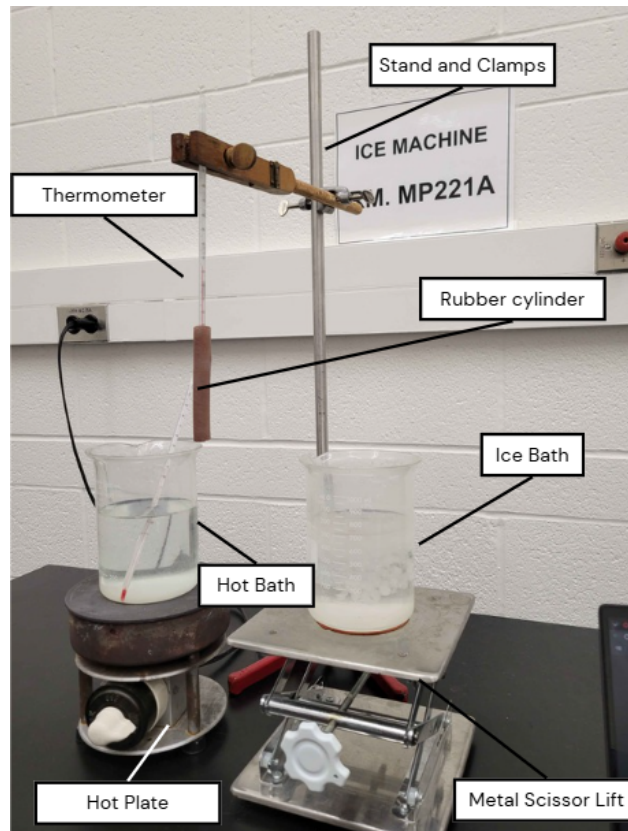


Figure 1: *Setup of the laboratory, hot bath on the left, ice bath on the right, note the water level is approximately equal, and the heights of the beakers are equal*

5. Insert the end of the thermometer into the rubber cylinder, clamp it onto the stand, ensure you can quickly move the thermometer from one beaker to another, but it **does not** touch the bottom or side of the beaker. The experimental setup should look like Figure 1.
6. Choose a period, τ and start a timer, place the "rubbered" thermometer into the cold bath. Every time half of the period has elapsed, place the "rubbered" thermometer into the unoccupied beaker and record the time Δt between when the thermometer reaches the new bath and when the temperature reaches a minimum (hot bath) or maximum (cold bath).
7. Repeat step 6 so that the "rubbered" thermometer was switched into the unoccupied beaker 6 times, and record the time delay for every switch.
8. Repeat steps 6 and 7 for different periods. For this iteration of the experiment, the periods chosen were 120, 180, 240, 300, 360, 420 and 480 seconds.

Raw Experimental Data

Radii	Value [m]
Inner Radius, r_1	$(3.60 \pm 0.25) \times 10^{-3}$
Outer Radius, r_2	$(8.05 \pm 0.25) \times 10^{-3}$

Table 1: Measured Inner and Outer Radii for Rubber material

For the following tables, the uncertainty of the time of the beaker switch and the maximum/minimum temperature reached are both instrumental uncertainties, and the uncertainty of the time delay was determined using Equation 10.

Time of Beaker Switch [± 5 s]	Time of Maximum/Minimum Temperature [± 1 s]	Time Delay [± 5 s]
60	101	41
120	158	38
180	218	38
240	277	37
300	339	39
360	399	39

Table 2: Time delay data when the alternation of the "rubbered" thermometer between the hot and cold water baths occurs with a period of 120 seconds.

Time of Beaker Switch [± 5 s]	Time of Maximum/Minimum Temperature [± 1 s]	Time Delay [± 5 s]
90	114	24
180	224	44
270	306	36
360	398	38
450	488	38
540	572	32

Table 3: Time delay data when the alternation of the "rubbered" thermometer between the hot and cold water baths occurs with a period of 180 seconds.

Time of Beaker Switch [± 5 s]	Time of Maximum/Minimum Temperature [± 1 s]	Time Delay [± 5 s]
120	156	36
240	273	33
360	395	35
480	514	34
600	634	34
720	755	35

Table 4: Time delay data when the alternation of the "rubbered" thermometer between the hot and cold water baths occurs with a period of 240 seconds.

Time of Beaker Switch [± 5 s]	Time of Maximum/Minimum Temperature [± 1 s]	Time Delay [± 5 s]
150	181	31
300	338	38
450	482	32
600	633	33
750	782	32
900	934	34

Table 5: Time delay data when the alternation of the "rubbered" thermometer between the hot and cold water baths occurs with a period of 300 seconds.

Time of Beaker Switch [± 5 s]	Time of Maximum/Minimum Temperature [± 1 s]	Time Delay [± 5 s]
180	210	30
360	389	29
540	571	31
720	750	30
900	930	30
1080	1109	29

Table 6: Time delay data when the alternation of the "rubbered" thermometer between the hot and cold water baths occurs with a period of 360 seconds.

Time of Beaker Switch [± 5 s]	Time of Maximum/Minimum Temperature [± 1 s]	Time Delay [± 5 s]
210	240	30
420	446	26
630	658	28
840	870	30
1050	1081	31
1260	1287	27

Table 7: Time delay data when the alternation of the "rubbered" thermometer between the hot and cold water baths occurs with a period of 420 seconds.

Time of Beaker Switch [± 5 s]	Time of Maximum/Minimum Temperature [± 1 s]	Time Delay [± 5 s]
240	156	22
480	273	29
720	395	31
960	514	31
1200	634	24
1480	755	28

Table 8: Time delay data when the alternation of the "rubbered" thermometer between the hot and cold water baths occurs with a period of 480 seconds.

Results and Analysis

In this lab, silicone rubber is used as the material, it has thermal properties given below [3]:

Property	Value [Units]
Thermal Conductivity, k	0.20 – 0.25 [W/mK]
Density, ρ	1100 – 1250 [kg/m^3]
Specific Heat Capacity, c_p	1050 – 1300 [J/kgK]

Table 9: Industrially accepted thermal properties of silicone rubber.

Applying Equation 2 and substituting values from Table 9, we obtain the general range of the theoretical thermal diffusivity for silicone rubber, m_{ind} , the rubber used in this experiment:

$$m_{ind} \approx 1.4 \times 10^{-7} \text{ to } 1.8 \times 10^{-7} [m^2/s] \quad (13)$$

For each period tested in the experiment, the average of the time delay Δt_{avg} is taken, and their uncertainties were calculated using Equation 8. The corresponding angular frequencies were calculated using Equations 11 and 12. The processed data is given in the table below:

$\omega [rad/s]$	$\pm [rad/s]$	$\Delta t [s]$	$\pm [s]$
0.052	0.002	38.2	5.2
0.035	0.001	37.6	6.7
0.0262	0.0006	34.2	5.2
0.0209	0.0004	33.8	5.7
0.0175	0.0002	29.8	5.2
0.0150	0.0002	28.4	5.5
0.0131	0.0001	28.6	5.9

Table 10: Data Table containing the angular frequency and the corresponding average time delay.

This data was curve-fitted to Equation 6 to generate Figure 2.

Overall, this fit seemed to be unsuitable for our collected data, this is further explained in the *error sources* below. The large χ^2 demonstrates that error sources are underestimated, or our model that assumes radial symmetry is inappropriate for the fit. Similarly, $R^2 \ll 0$ again indicates an unsuitable fit in our case. Finally, as seen from the function curve and the residual data, our data crosses the fit function discontinuity. Thus, it demonstrates an overall linear increasing residual trend, with smaller ω values relating to a negative residual, bigger ω values relating to a positive residual.

From the resulting fit, the experimental value of thermal diffusivity is:

$$m_{exp} = (4.2 \pm 0.3) \times 10^{-8} [m^2/s] \quad (14)$$

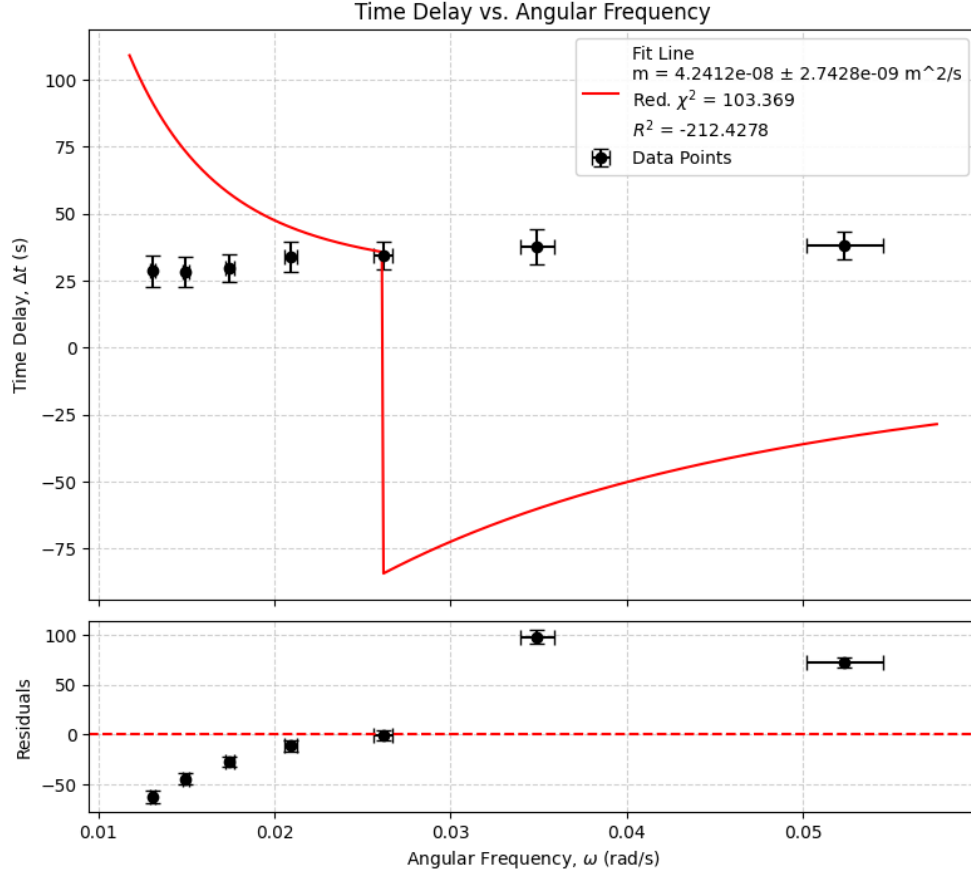


Figure 2: Upper chart: Δt as a function of ω , data points are black and the fit function is red. Note the discontinuity at $\omega \approx 0.026$. The Reduced Chi Squared is $\chi^2 = 103.369$, and the coefficient of determination is $R^2 = -212.4278$. Lower chart: Residuals of the fit.

The value m_{exp} is approximately 3 times smaller than m_{ind} .

Since m represents the ratio of the heat conducted through the material to the heat stored per unit volume [4], this indicates that in silicone rubber, most of the heat is absorbed by the material and only a small portion of heat is allowed to further diffuse in a given time period, which is the causation of the time delay in this experiment. In addition, a smaller experimental result indicates that our lab material takes a greater amount of time for heat to diffuse through than the ideal silicone rubber.

However, this experiment was affected by sources of error. One of the sources of error is the varied transferal procedure between water baths. All calculations in this experiment assumes an instantaneous transferal between water baths, which is physically impossible. This introduces noise for ω and Δt collected and inconsistency for the temperature gradient for the exposure to room temperature.

In addition, the non-uniformity of the material also attributes to the error. The varying

radius around the thermometer induces non-uniform heat absorption along the radial axis, which is unaccounted for in our symmetrical idealization. Similarly, the non-uniform thermal contact resistance (air gaps) between the thermometer and the rubber is neglected, this adds an extra layer of varying resistance, despite it sharing a similar low thermal conductivity with rubber.

Conclusion

Overall, this experiment demonstrated the time delay for heat to diffuse through materials, in this case silicone rubber. A thermal diffusivity for silicone rubber was determined through a best fit, $m_{exp} = (4.2 \pm 0.3) \times 10^{-8} [m^2/s]$, which is approximately 3 times smaller than the industrially accepted range. Even though the fit was unsuitable, the experiment still demonstrated a reasonable order of magnitude and reinforcing the fact that heat mitigates slowly in silicone rubber.

References

- [1] Philip R. Bevington and D. Keith Robinson. *Data Reduction and Error Analysis for the Physical Sciences*. McGraw-Hill, 3rd edition, 2003.
- [2] Joint Committee for Guides in Metrology (JCGM). *Evaluation of measurement data — Guide to the expression of uncertainty in measurement (JCGM 100:2008)*. ISO, https://www.iso.org/sites/JCGM/GUM/JCGM100/C045315e-html/C045315e_FILES/MAIN_C045315e/05_e.html, 2008.
- [3] CHEMnetBASE. *CRC Handbook of Chemistry and Physics - Chemical Search*. <https://hbcpc.chemnetbase.com/chemical/ChemicalSearch.xhtml?dswid=5565&faces-redirect=true>.
- [4] Yunus A. Çengel, John M. Cimbala, and Robert H. Turner. *Fundamentals of Thermal-Fluid Sciences*. McGraw-Hill, 5th edition, 2017.